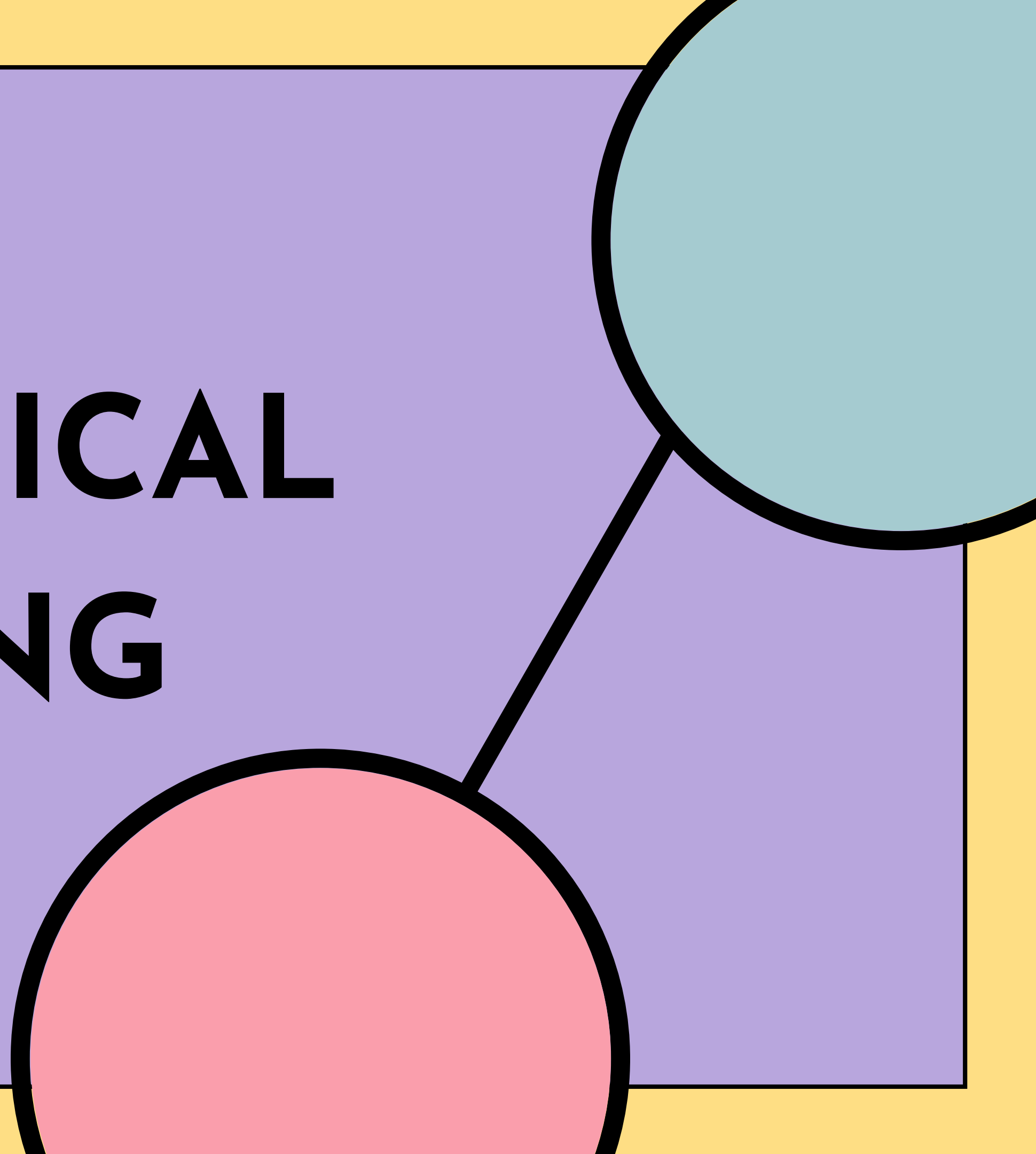


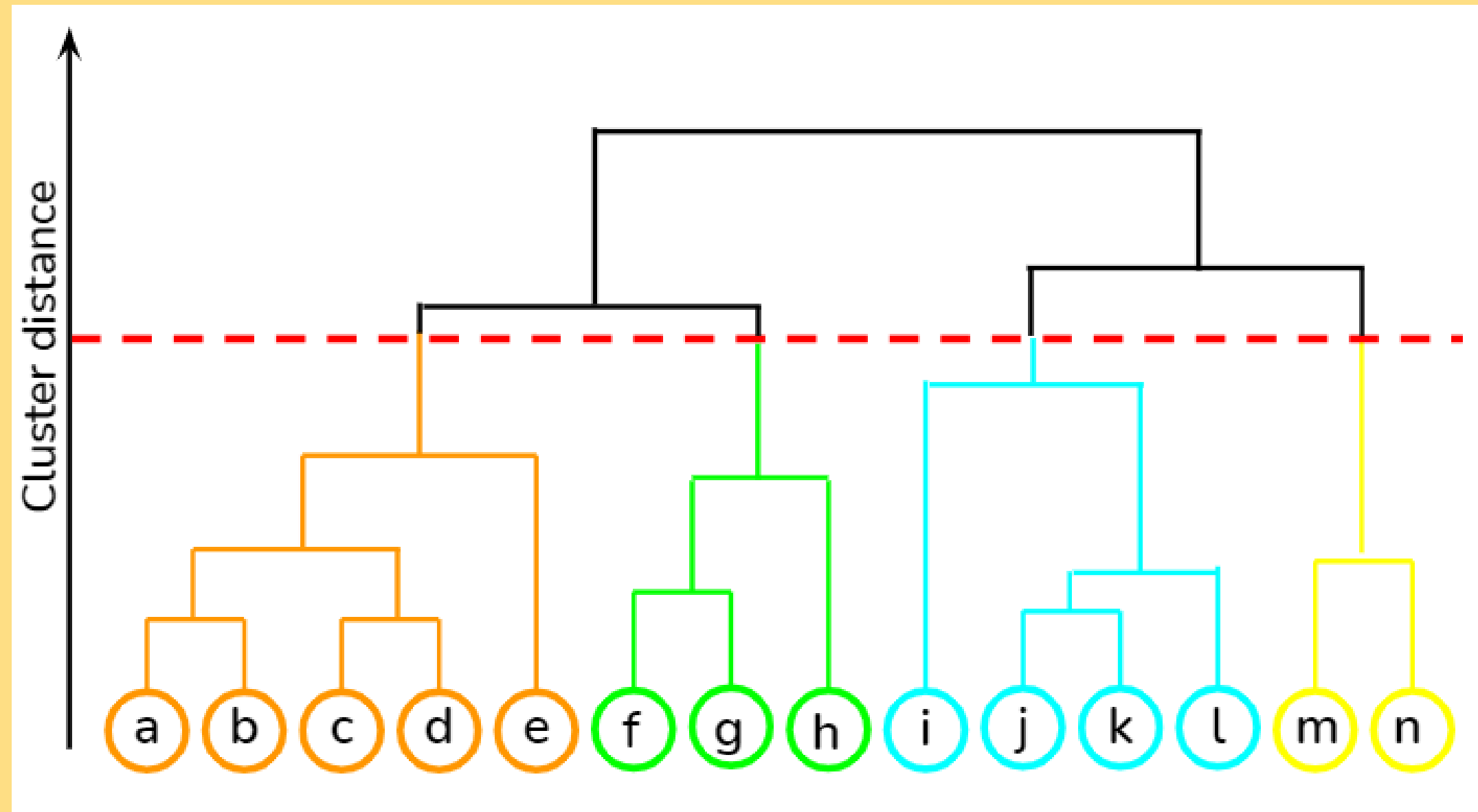
HIERARCHICAL CLUSTERING



What is Hierarchical Clustering?

- It is a clustering method that creates a hierarchical tree of objects to be clustered hierarchically.
- The tree represents the relationships between objects and shows how objects are clustered at different levels.

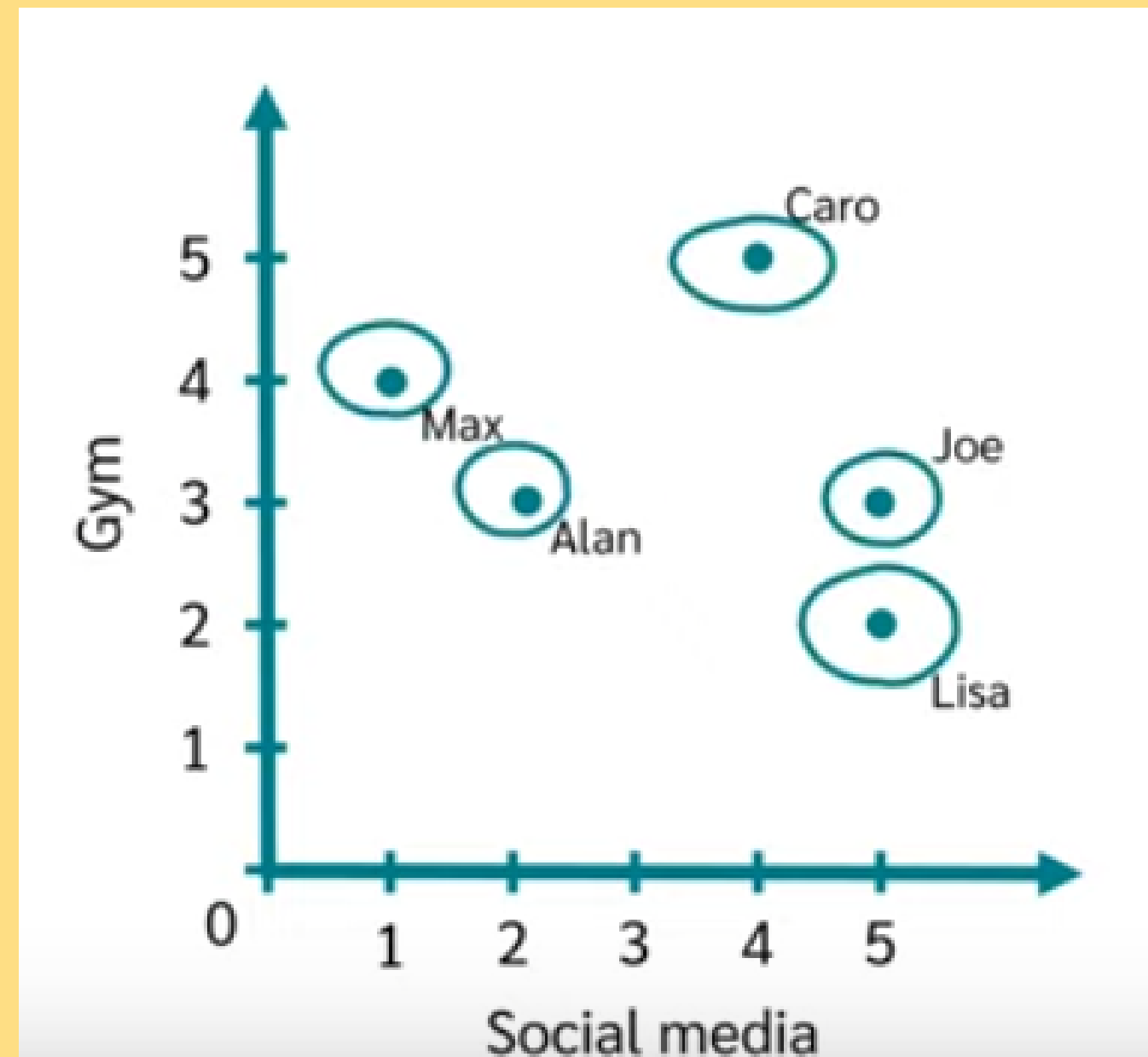
What is Hierarchical Clustering?



Dendrogram

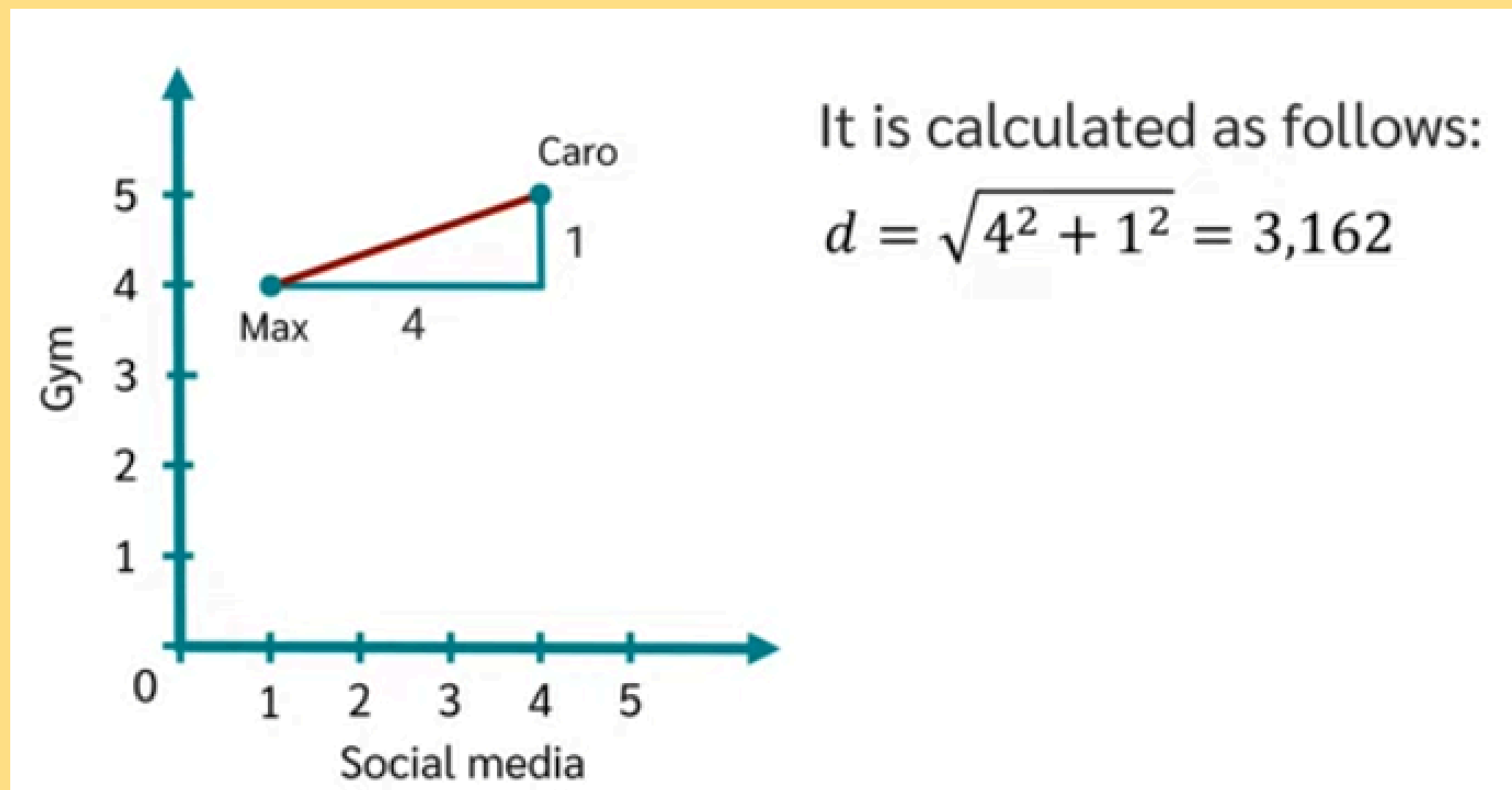
**How does Hierarchical
Clustering work?**

**1- Assign a cluster to each individual point
(number of clusters = number of points)**



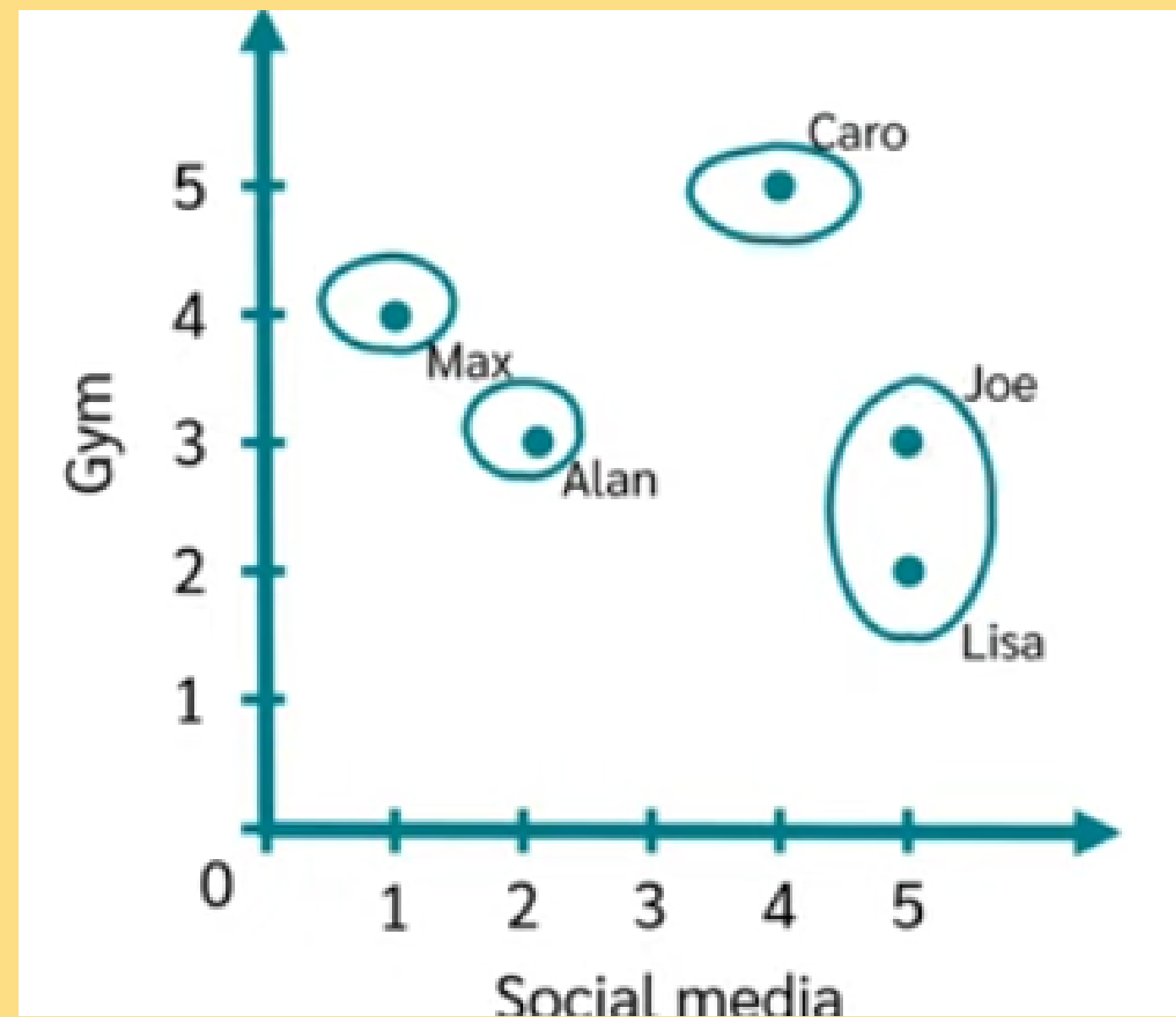
2- Calculate Pairwise distances

- Compute the distance between every pair of clusters using a distance metric .



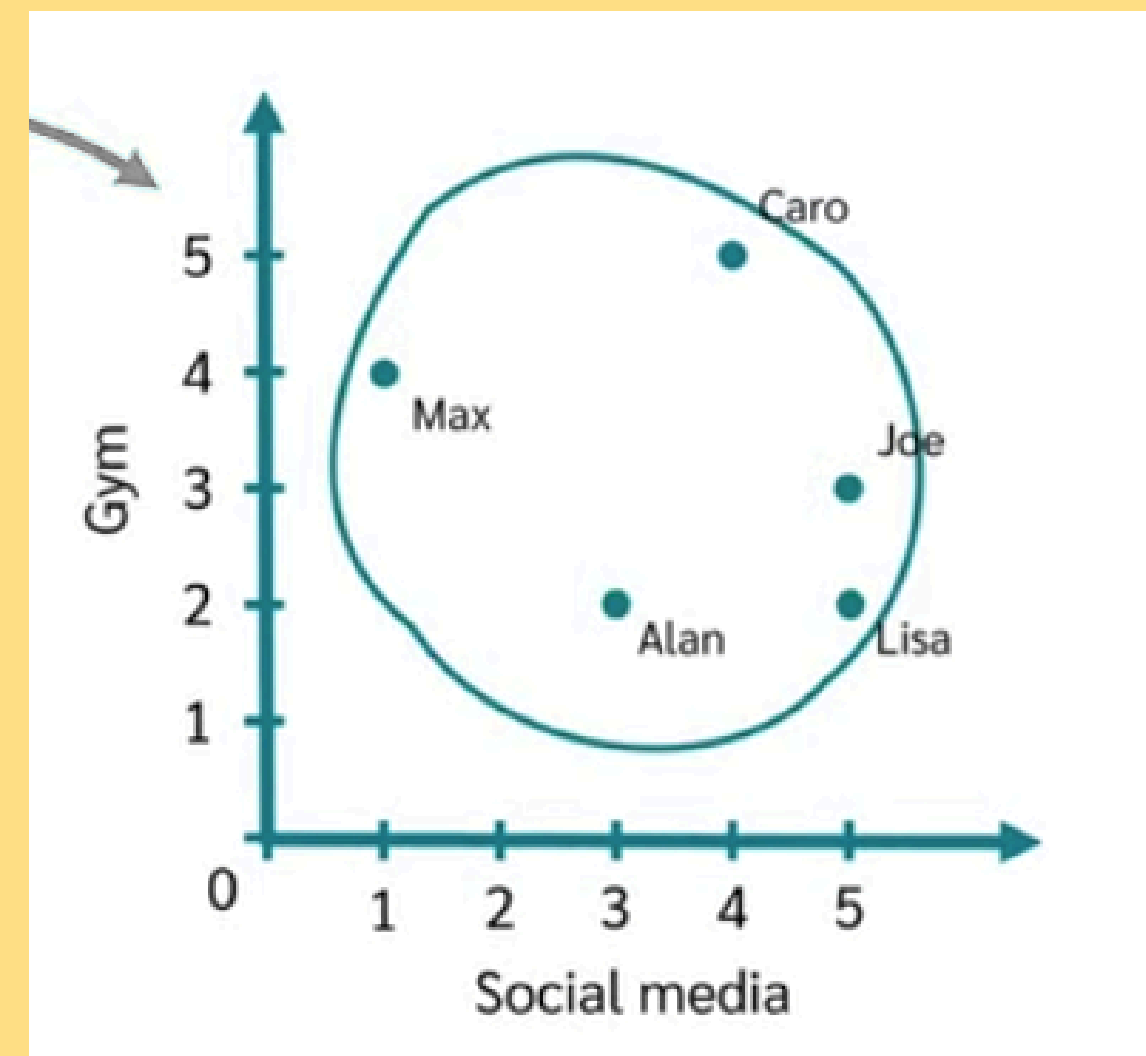
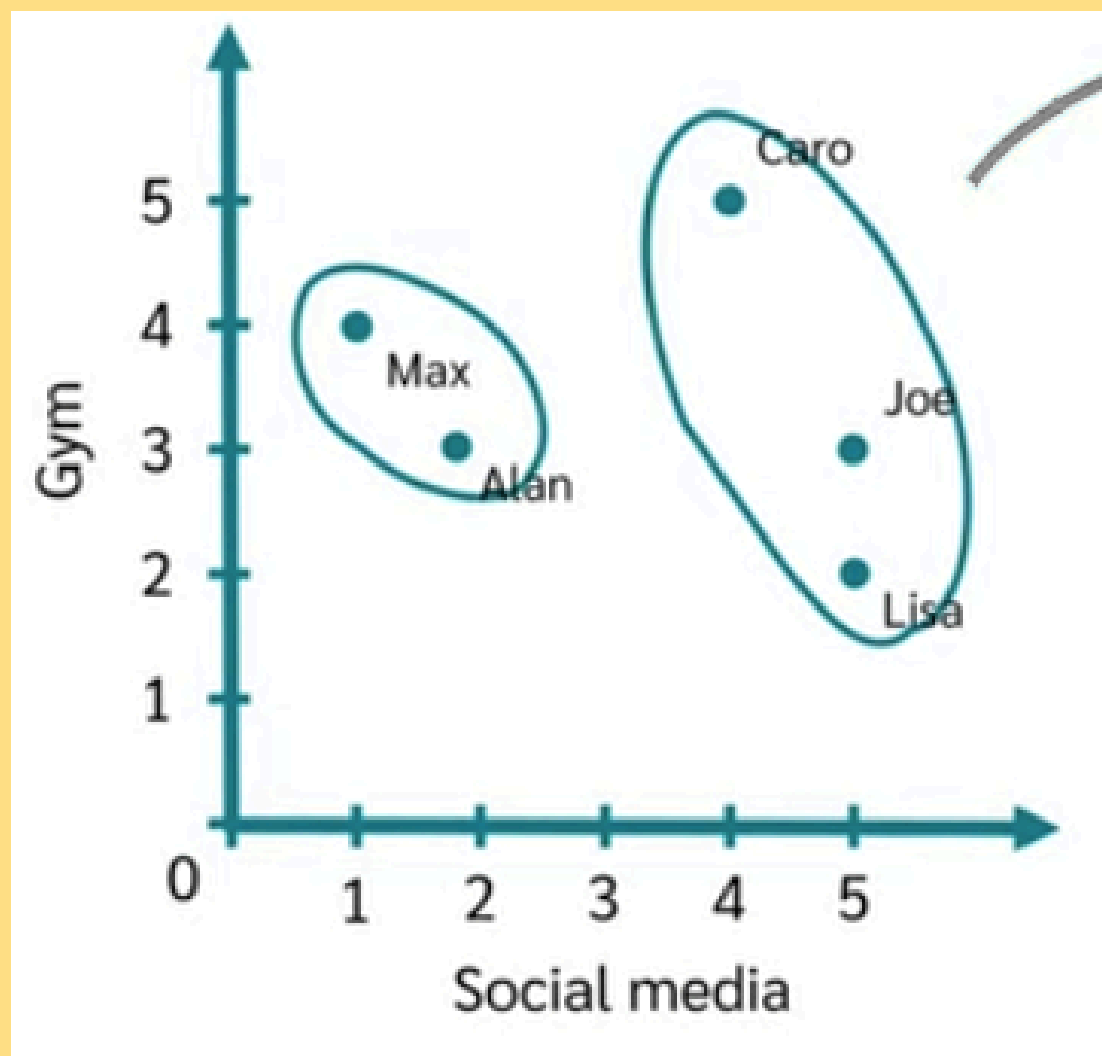
3- Link clusters

- Identify the two clusters that are closest to each other based on the distance metric, and merge these two clusters into a single cluster .



4- Repeat the process

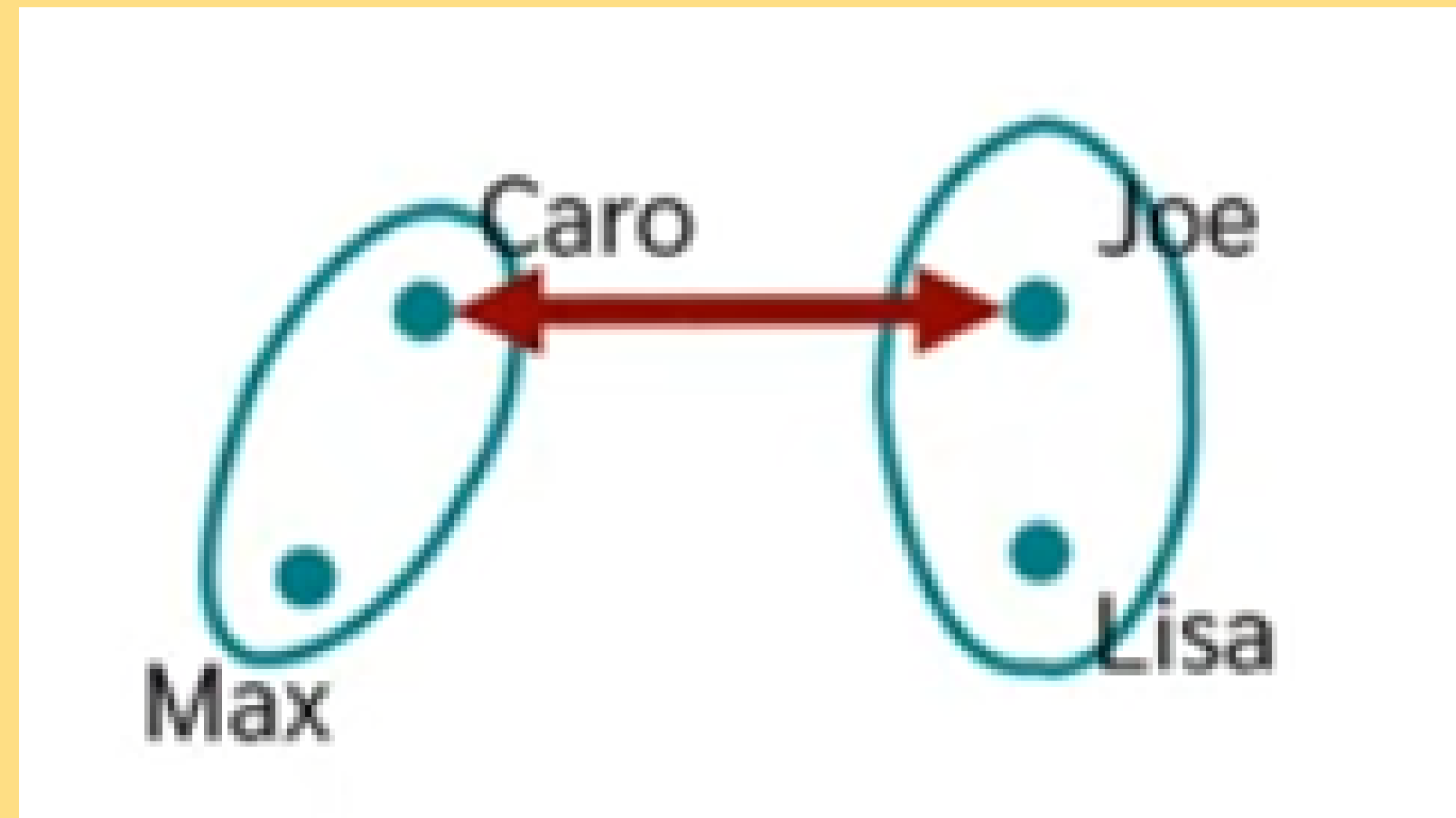
- Continue merging the closest clusters and updating the distance metric until all data points are merged into a **single cluster**, or until the desired number of clusters is reached.



**How to link cluster with
more than one point?**

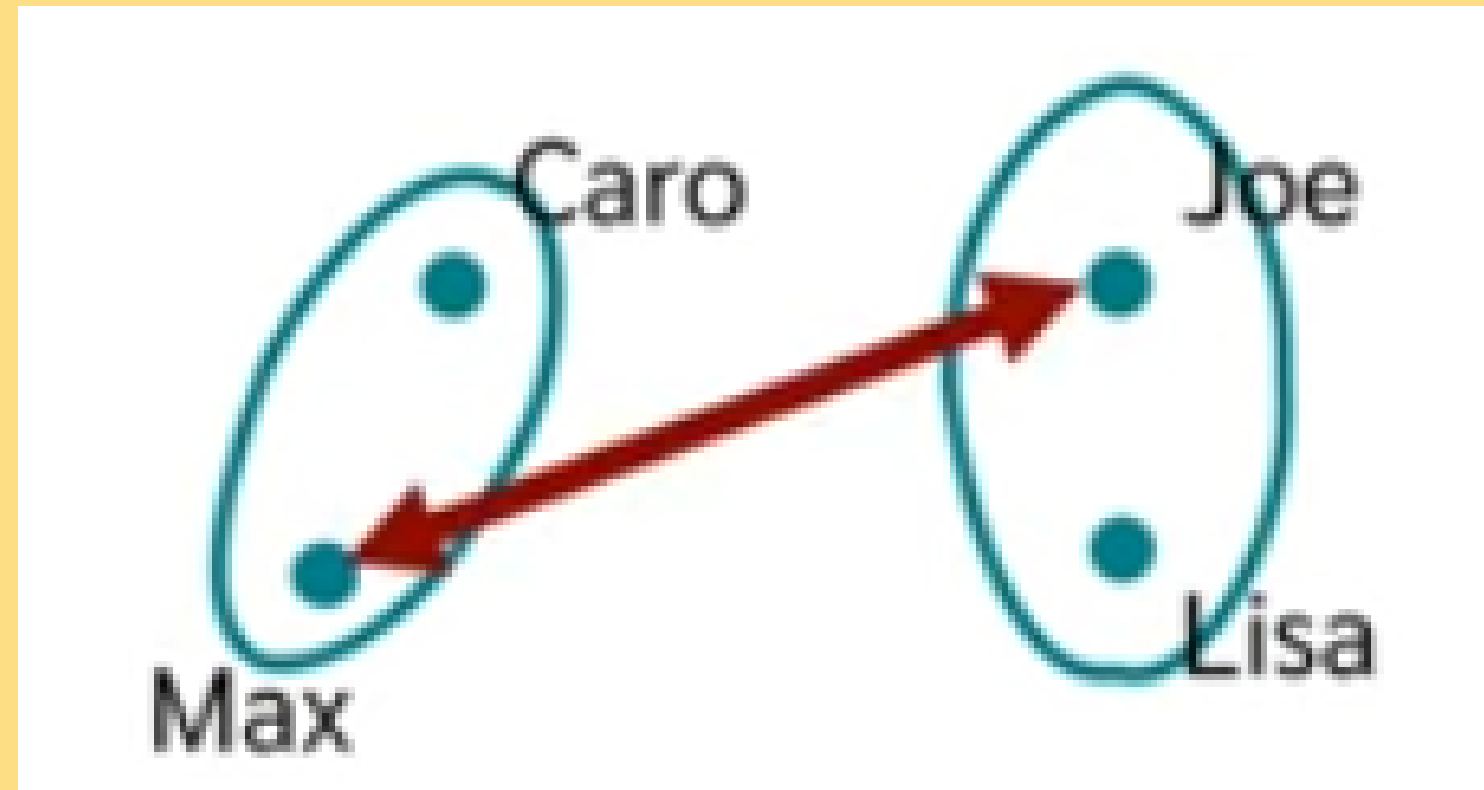
1- Single-linkage

- uses the distance between the **closest** elements in the cluster



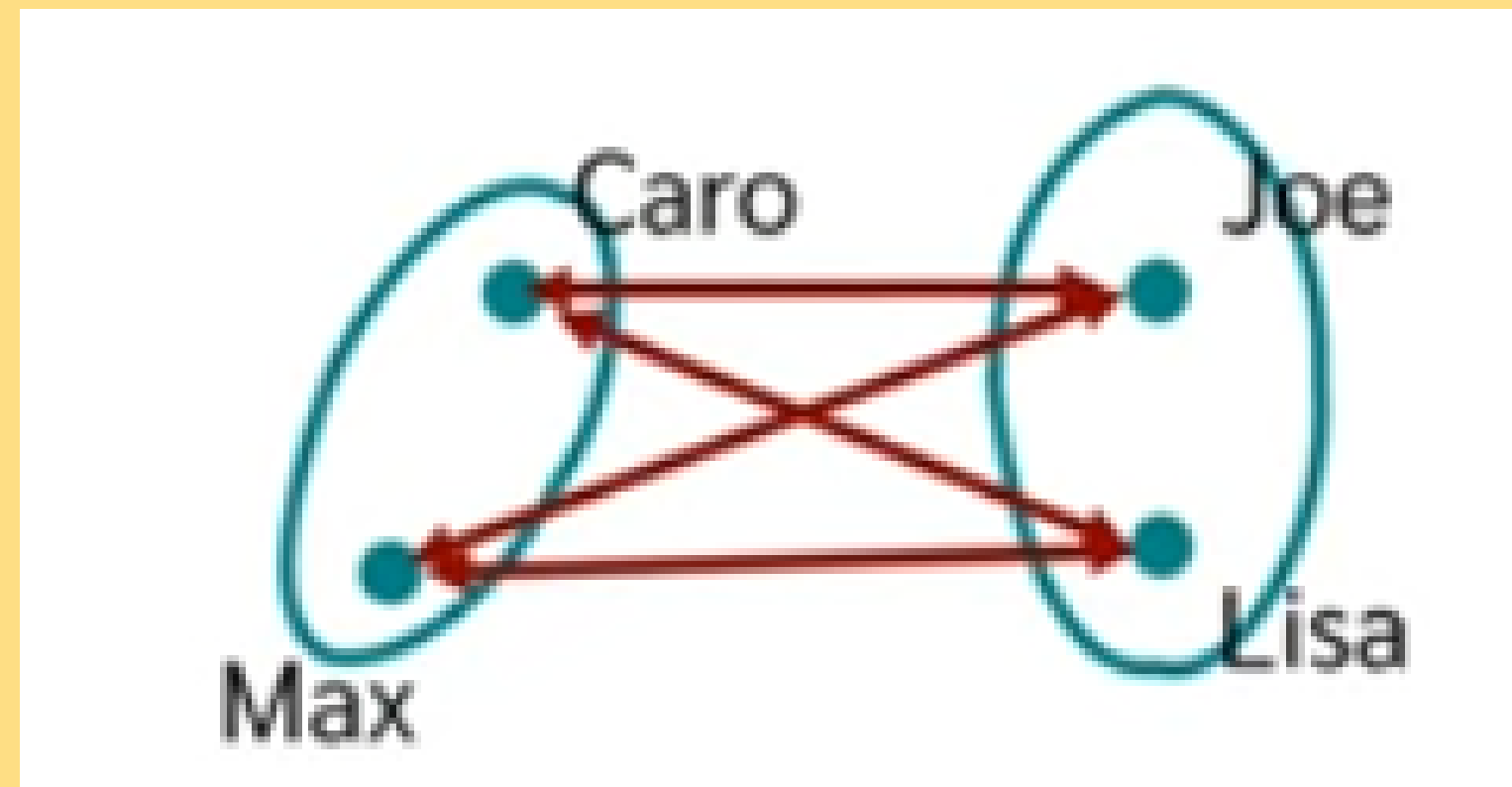
2- Complete-linkage

- uses the distance between the **most remote** elements of the cluster.



2- Average-linkage

- uses the average of all pairwise distances.

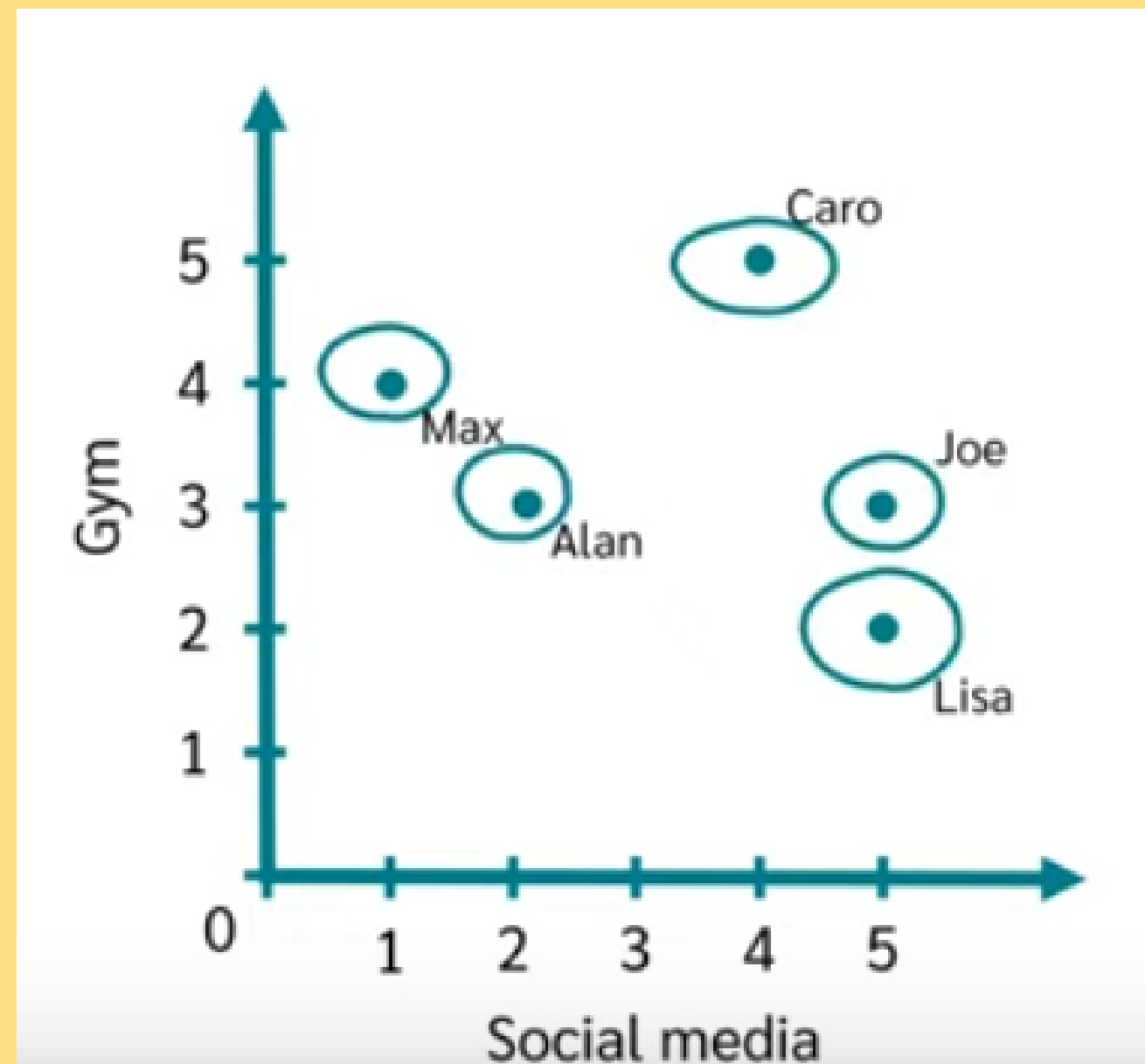


From each combination, the distance is calculated, and then we compute the average of all distances

Full Example

1- Every single point is a cluster

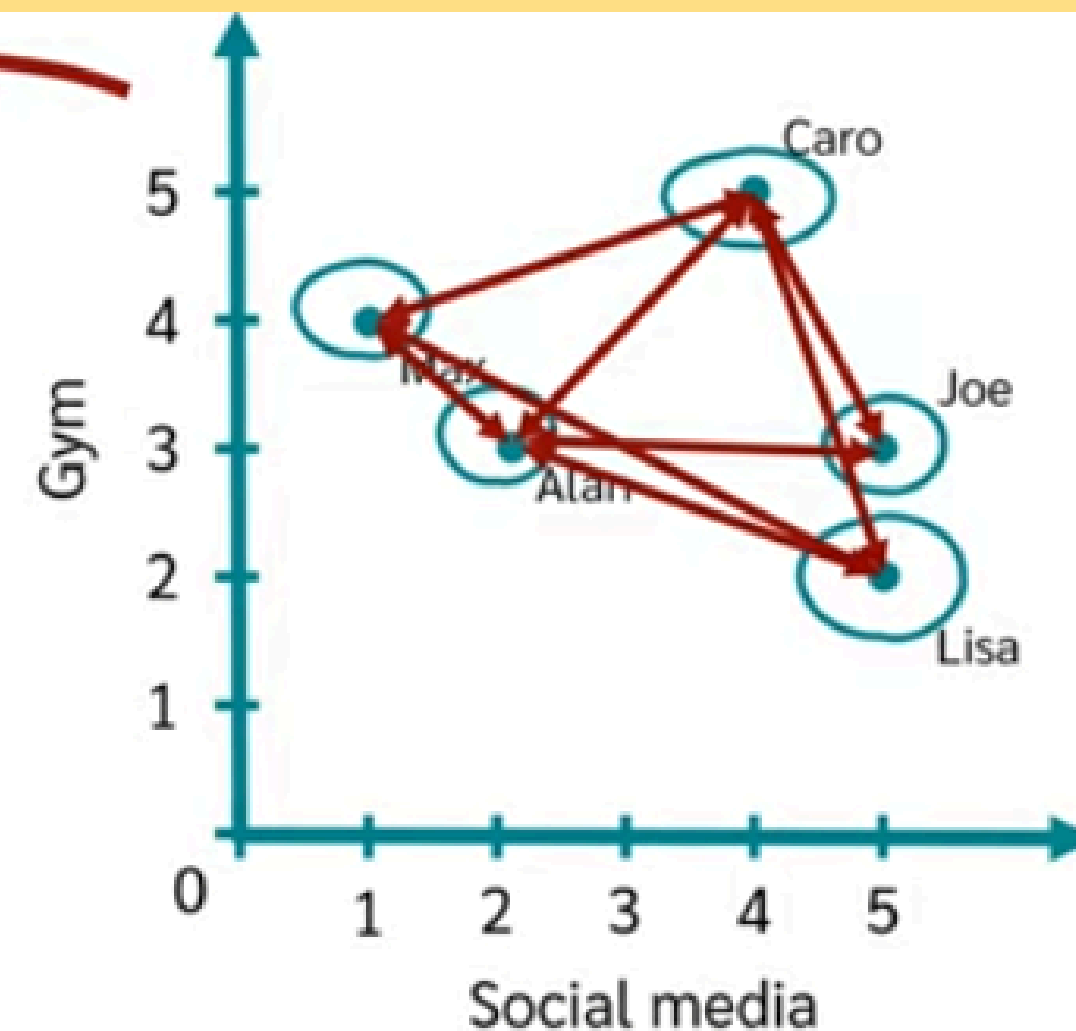
- This is the dataset we are going to work on; we created a cluster for each point.



2- Measure the distance between each cluster

- We create a distance matrix that contains the distance between each cluster and the other clusters.

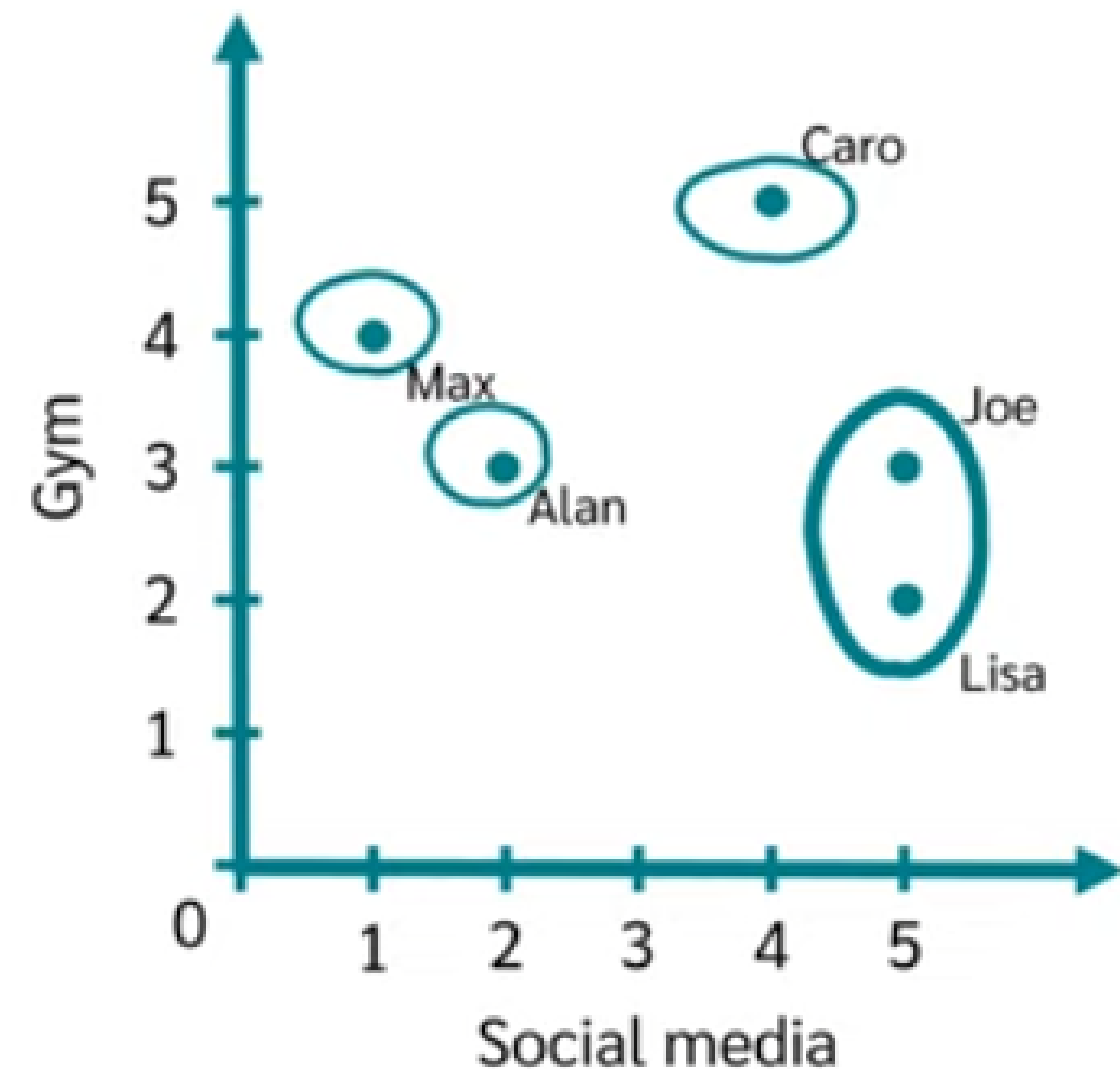
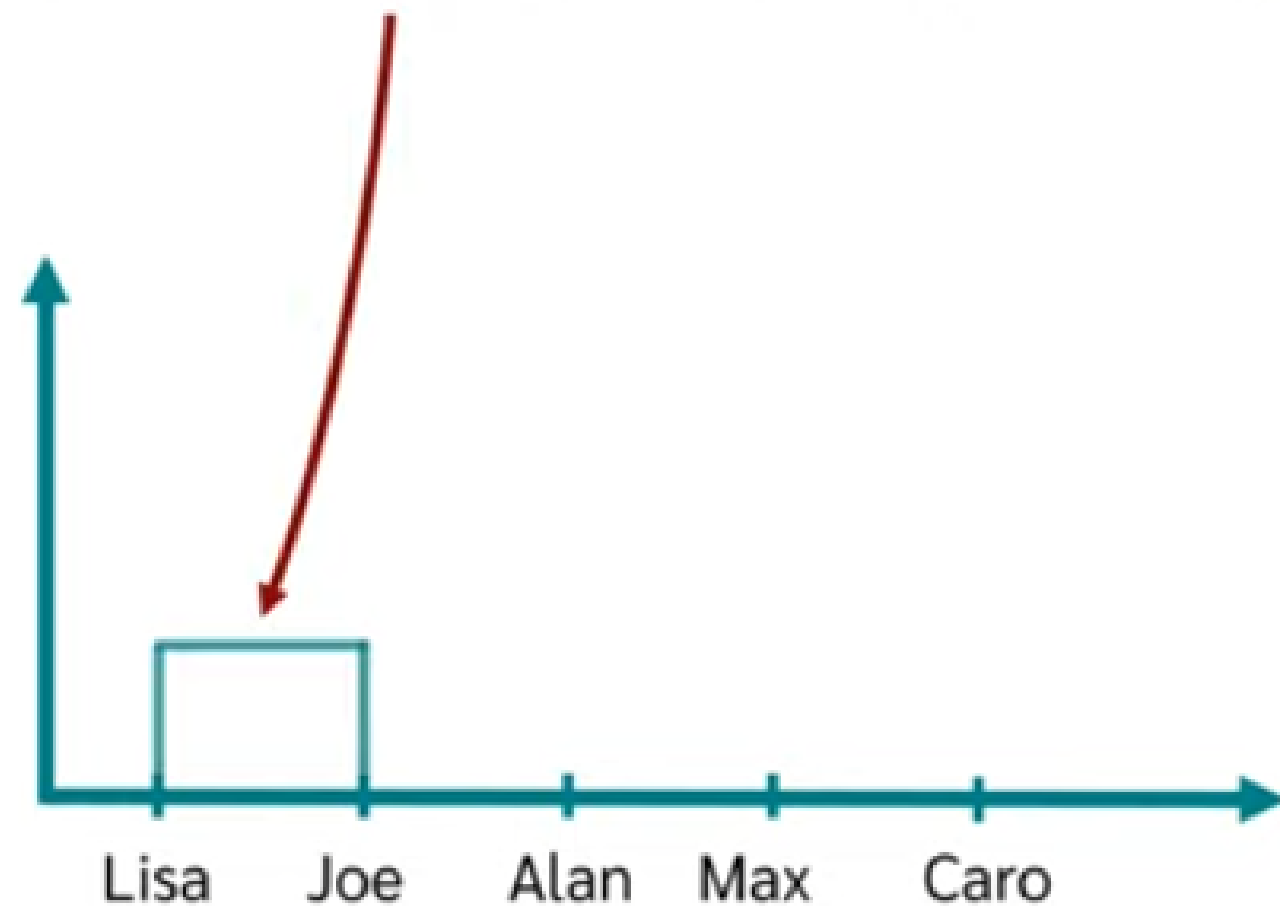
	Alan	Lisa	Joe	Max	Caro
Alan	0				
Lisa	3,16	0			
Joe	3,00	1,00	0		
Max	1,41	4,47	4,12	0	
Caro	2,83	3,16	2,24	3,16	0



3- Merge the closest clusters

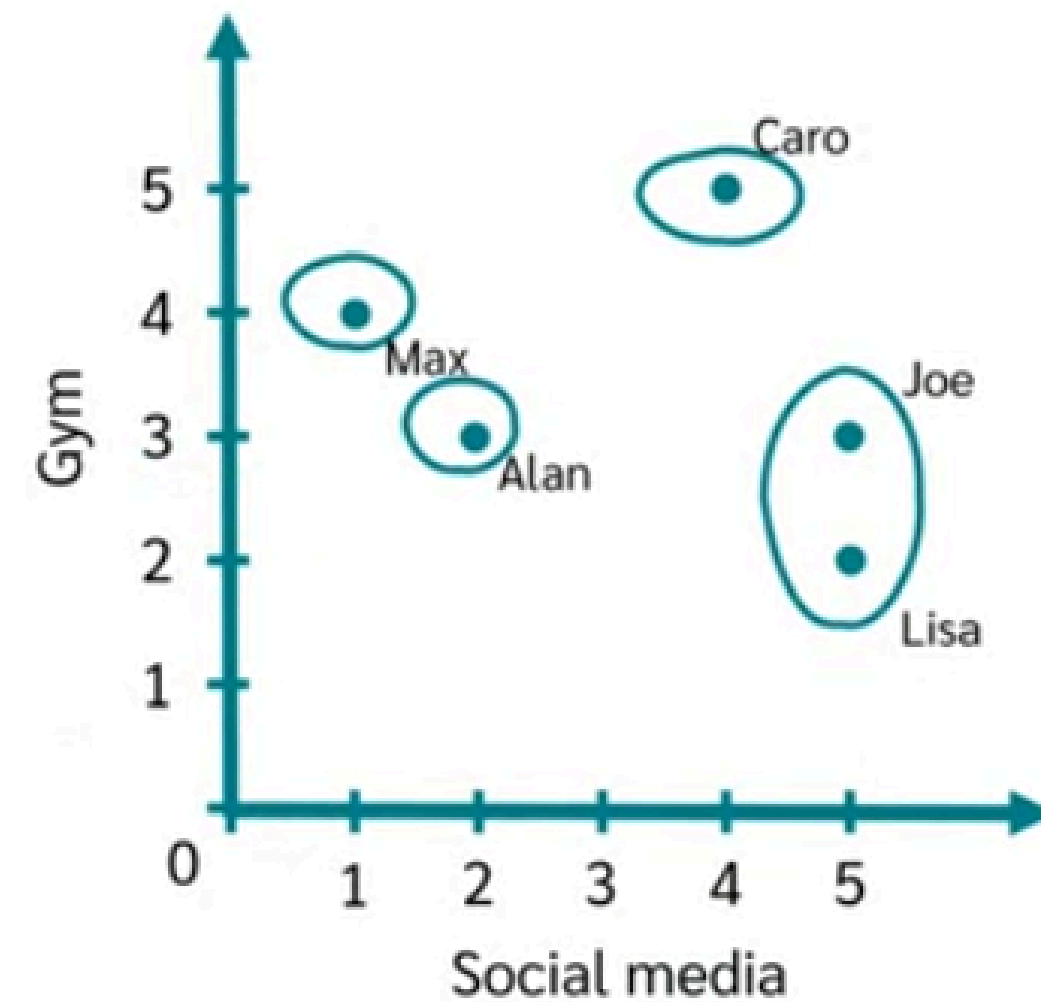
- Lisa and Joe are the closest clusters with a distance of 1.0

In our **tree diagram** or **dendrogram** we can draw the first connection.



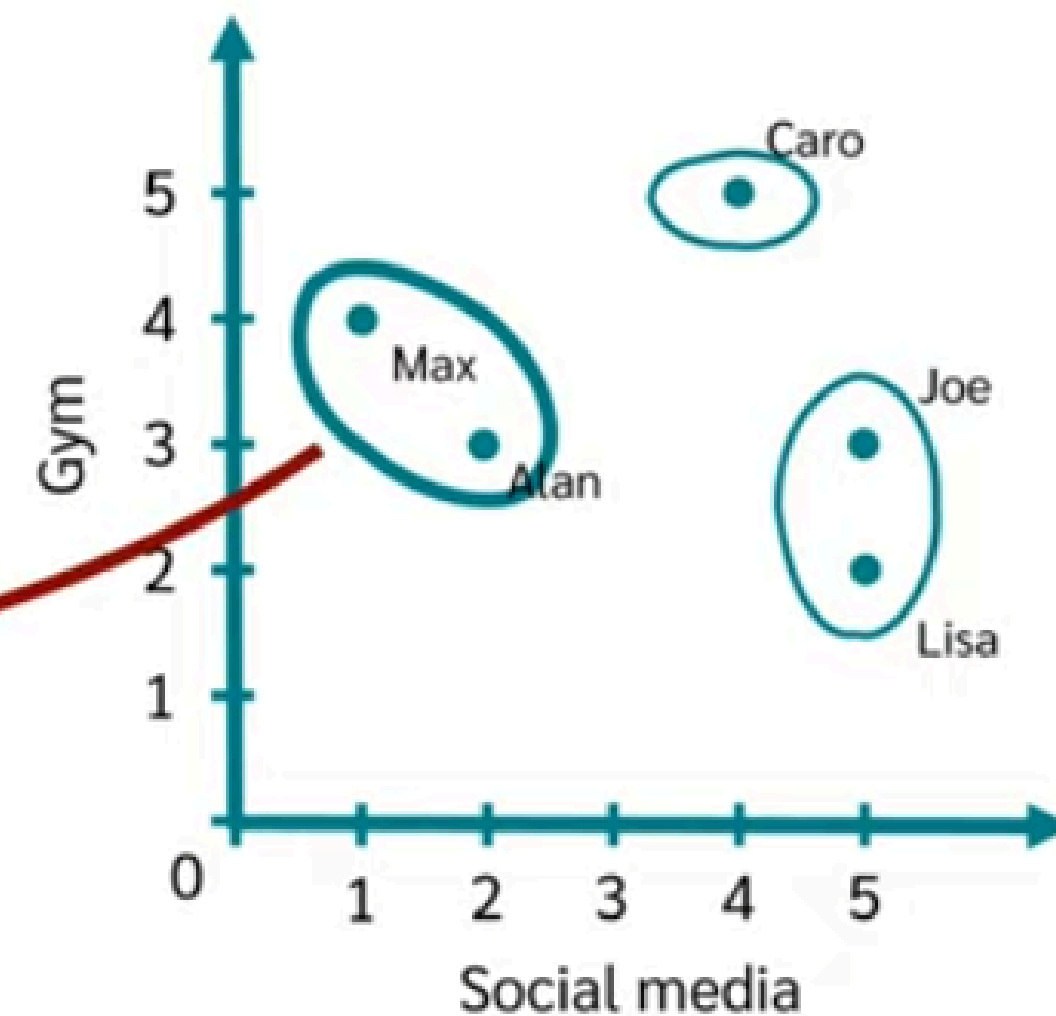
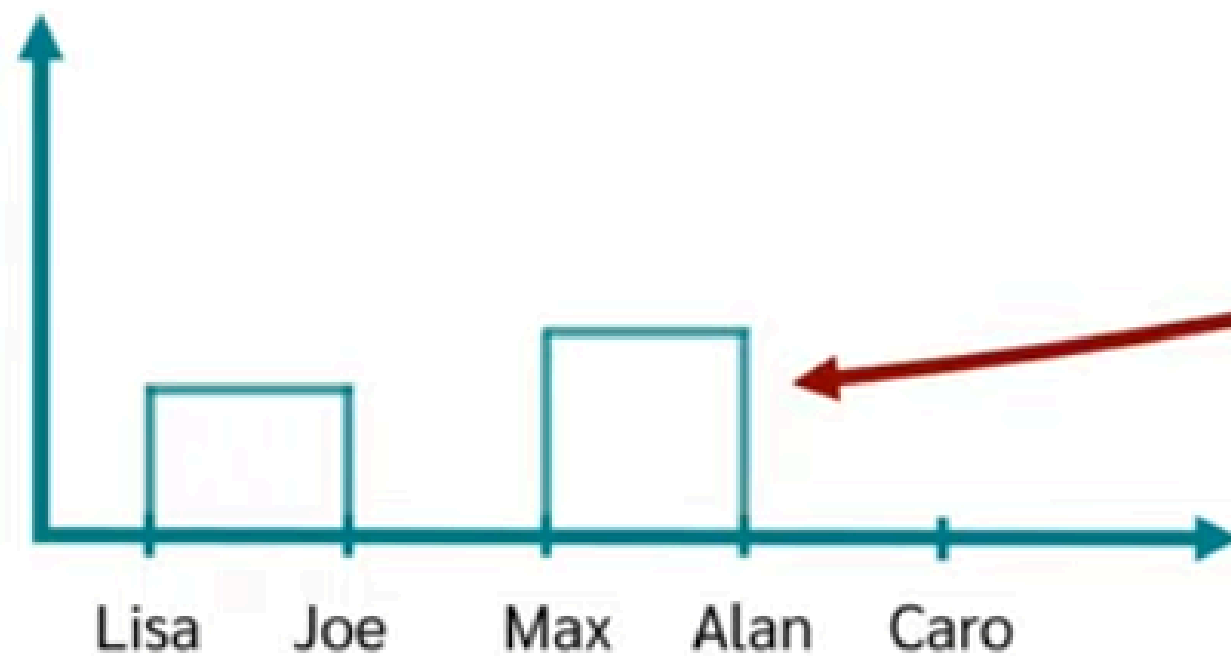
4- Update the distance matrix

	Alan	Lisa, Joe	Max	Caro
Alan	0			
Lisa, Joe	3,00	0		
Max	1,41	4,12	0	
Caro	2,83	2,24	3,16	0

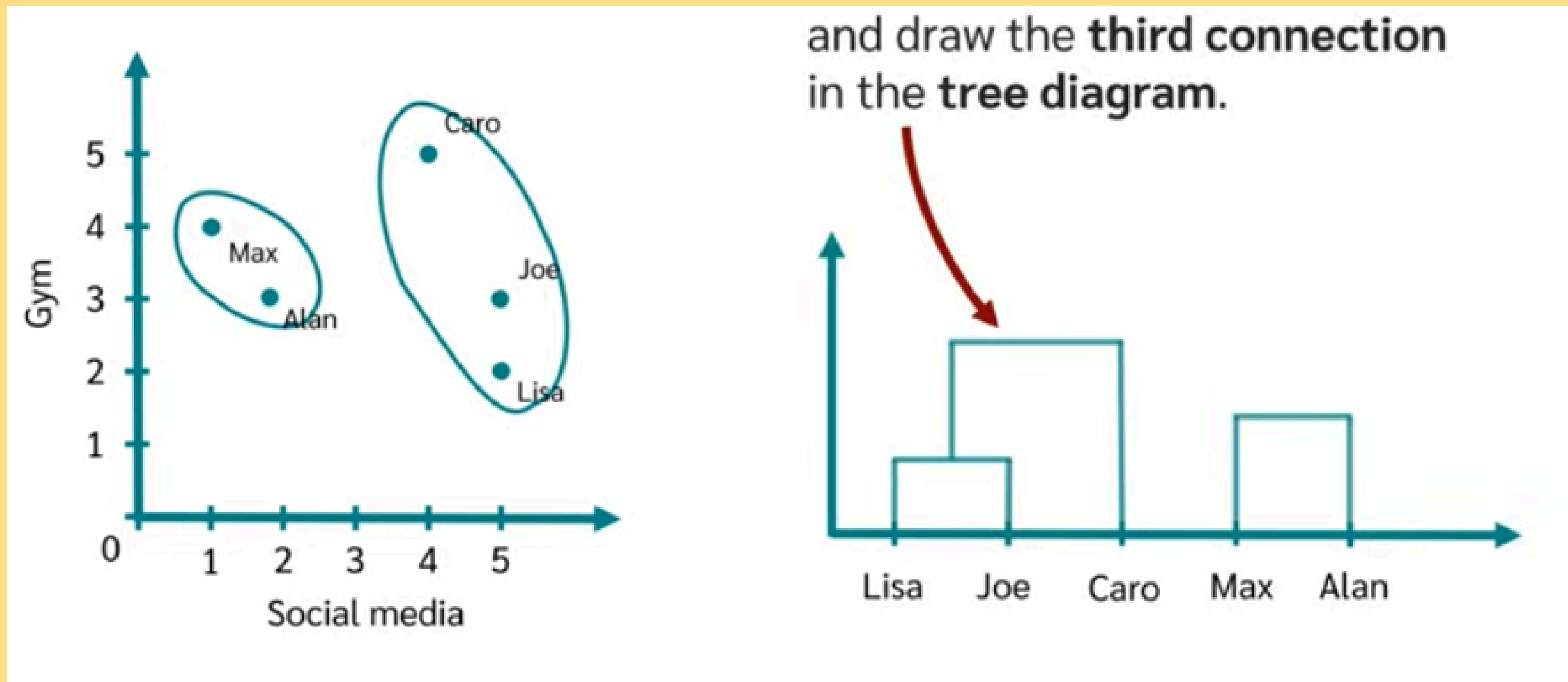


5- Choose the new closest clusters

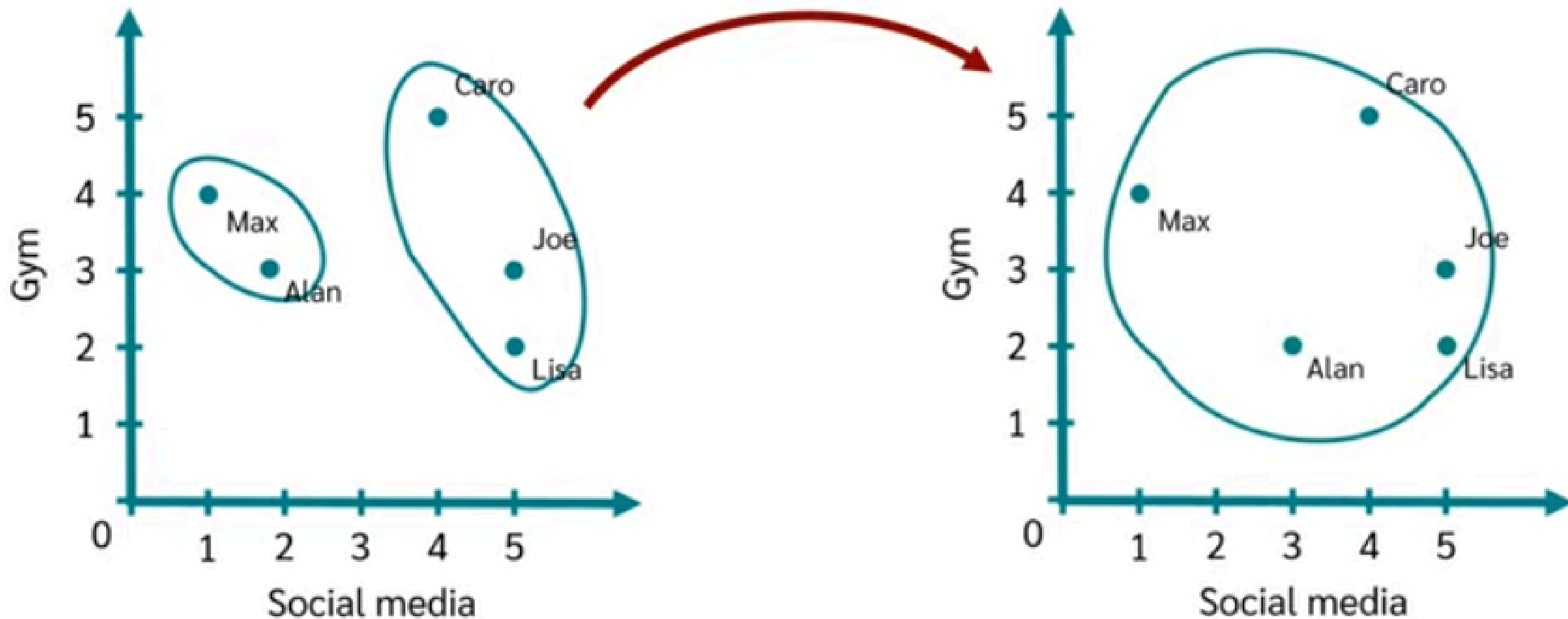
In our **tree diagram** or **dendrogram** we can draw the second connection.



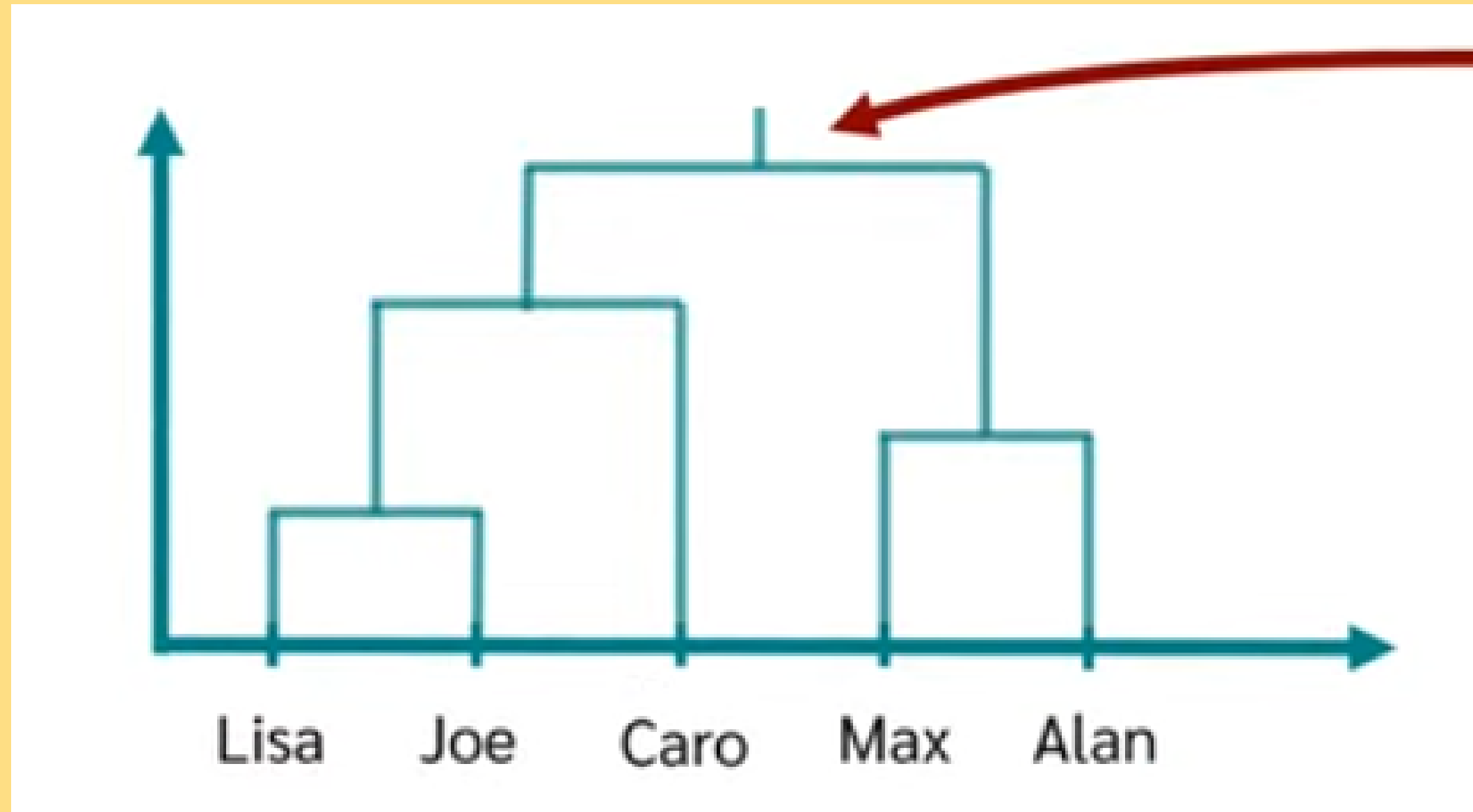
6- Continue the same steps until we reach our goal



6- Continue the same steps until we reach our goal



7- Final Clusters



K-means vs Hierarchical

Aspect	K-means	Hierarchical
Complexity	faster	Slower, requires more iterations
Scalability	better with large datasets	Not preferable with large datasets
Parameters	Require (K) before clustering	Does not require parameters
Output	It produces one clustering scheme	It produces a hierarchy of clusters, which can be cut at different levels to obtain different numbers of clusters

When to use which?

When to use..

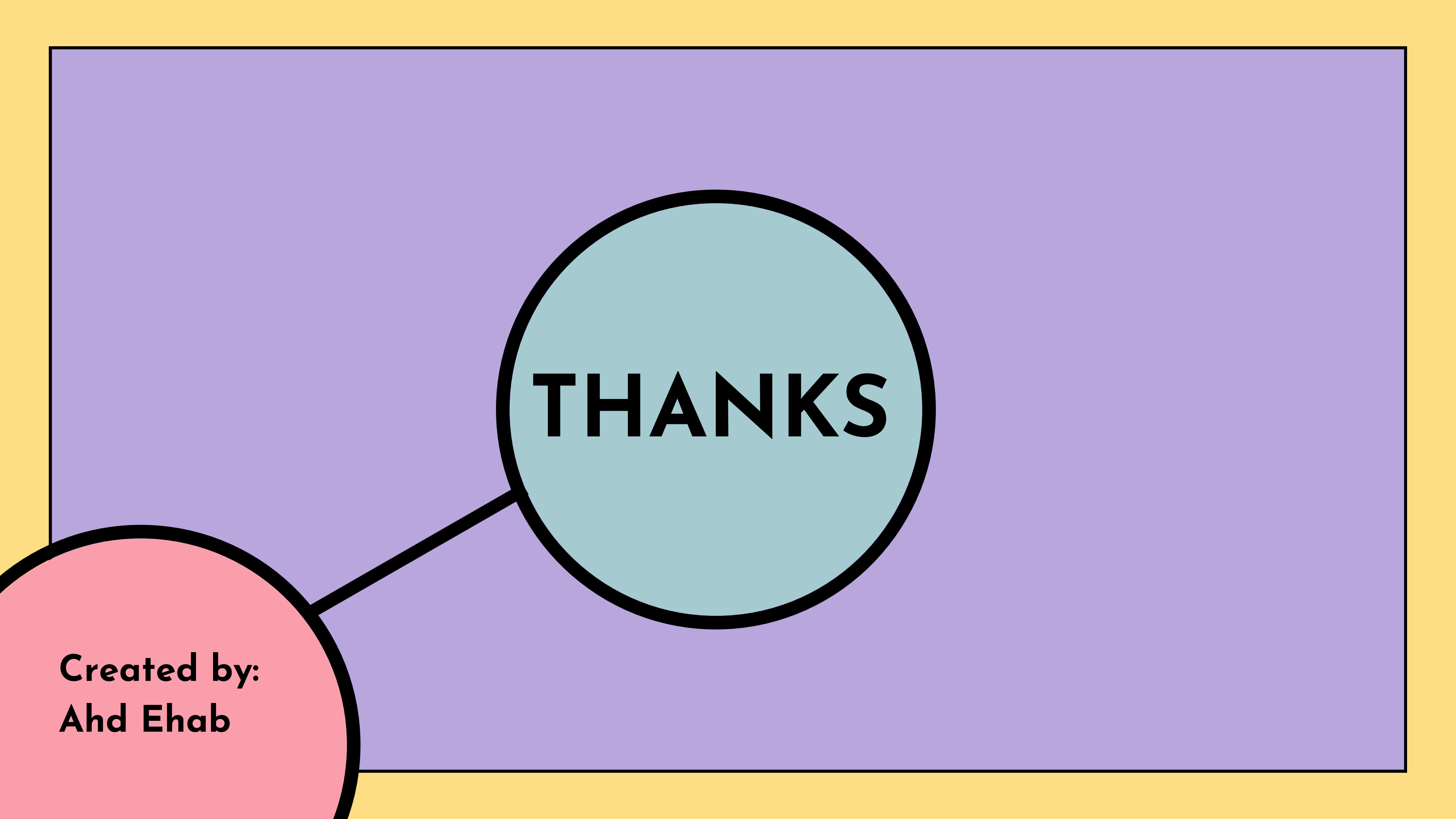
K-means:

- Best for large datasets
- When the number of clusters is known
- When clusters are roughly spherical

Hierarchical

- Good for smaller datasets
- When you want to explore the data to determine the number of clusters
- Better when the dataset is imbalanced

Any Questions?



THANKS

**Created by:
Ahd Ehab**